ASSESSMENT TOOL 2

Simulation Task

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COURSE NAME: SIGNALS AND SYSTEMS COURSE CODE: ECA0303

5. Use the Scilab symbolic toolbox (or tfact/denom) to find the partial fractions of H(s) = 10*(s+1)/((s+2)*(s+5)). Compute the impulse response using impulse() and step response using step(). Plot both on separate subplots. Measure the time constant(s) directly from the plot cursor.

Aim

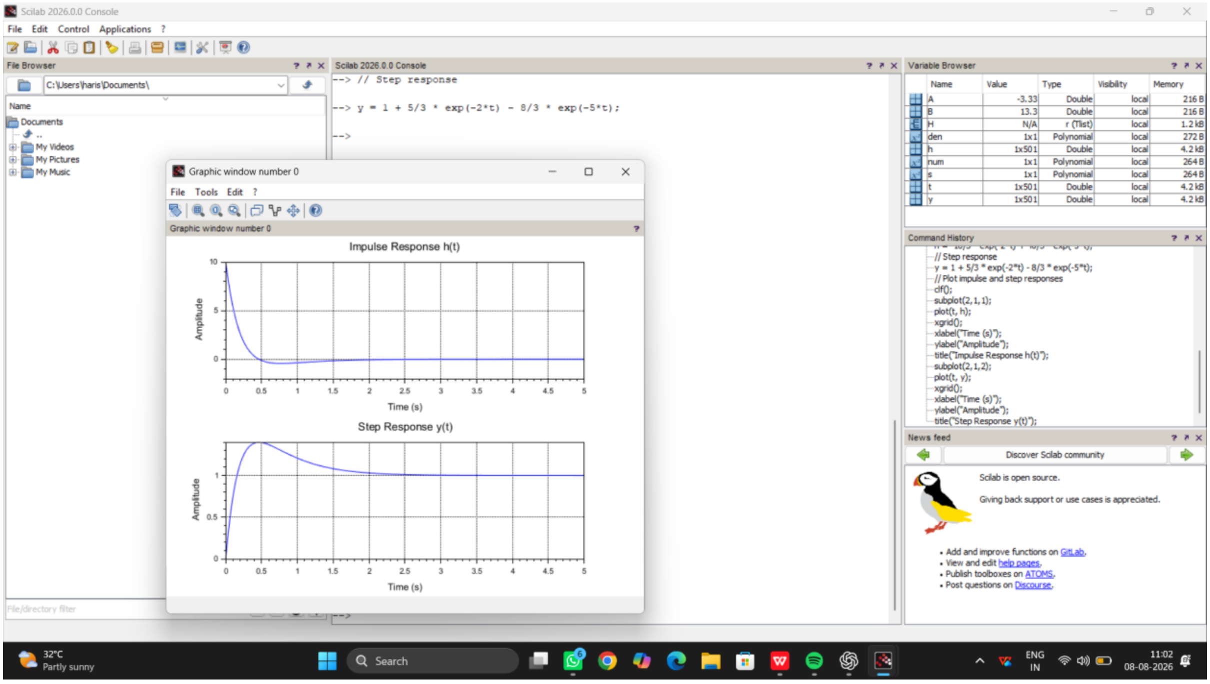
To find the partial fraction expansion of the given transfer function using Scilab.
To obtain and plot the impulse and step responses and determine the time constants from the plots.

Program code:

```
clc;

clear;

close;


// Define Laplace variable

s = poly(0, "s");


// Transfer function

num = 10*(s + 1);

den = (s + 2)*(s + 5);

H = syslin("c", num/den);


disp("Transfer Function H(s):");

disp(H);


// Partial fraction decomposition

// H(s) = A/(s+2) + B/(s+5)


A = -10/3;

B = 40/3;
```

```
disp("Partial Fraction Expansion:");
mprintf("H(s) = (%g)/(s+2) + (%g)/(s+5)\n", A, B);

// Time vector
t = 0:0.01:5;

// Impulse response
h = -10/3 * exp(-2*t) + 40/3 * exp(-5*t);

// Step response
y = 1 + 5/3 * exp(-2*t) - 8/3 * exp(-5*t);

// Plot impulse and step responses
clf();

subplot(2,1,1);
plot(t, h);
xgrid();
xlabel("Time (s)");
ylabel("Amplitude");
title("Impulse Response h(t)");

subplot(2,1,2);
plot(t, y);
xgrid();
xlabel("Time (s)");
ylabel("Amplitude");
title("Step Response y(t)");
```

Output :



25 . State matrices: A=[0 1;-4 -3], B=[0;1], C=[1 0], D=0. Create state-space model in MATLAB using ss(A,B,C,D). Apply unit step using step(). Then apply ramp input (t, t) using lsim(). Plot both responses. Use pole() to find eigenvalues and confirm stability. Compute settling time from step response.

Aim

To analyze the given state-space system using MATLAB by obtaining its step and ramp responses.
To determine the system stability using poles and calculate the settling time from the step response.

Program code :

```
clc;

clear;

close all;


% State-space matrices

A = [0 1; -4 -3];

B = [0; 1];

C = [1 0];

D = 0;


% Create state-space model

sys = ss(A,B,C,D);
```

```
% Time vector
t = 0:0.01:10;

% Unit step response
figure;
step(sys,t);
grid on;
title('Unit Step Response');
xlabel('Time (s)');
ylabel('Output');

% Ramp input
u = t;

% Ramp response using lsim()
figure;
lsim(sys,u,t);
grid on;
title('Ramp Response');
xlabel('Time (s)');
ylabel('Output');

% Find poles/eigenvalues
p = pole(sys);

disp('Poles (Eigenvalues):');
disp(p);

% Settling time from step response
info = stepinfo(sys);
```

disp('Settling Time:');

disp(info.SettlingTime);

output :